

CH32X3x5 Evaluation Board Reference

Version: V1.0

<https://wch-ic.com>

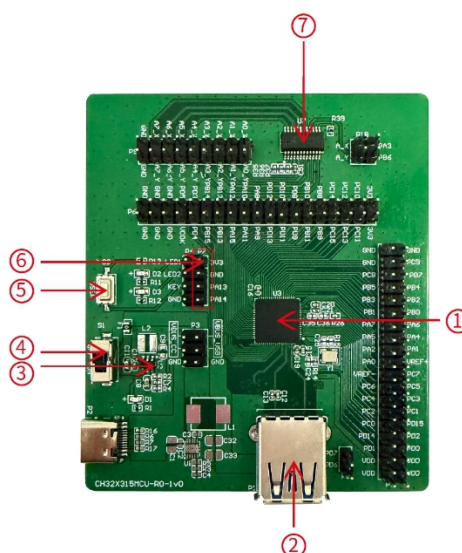
1. Overview

This evaluation board is applied to the development of the CH32X3x5 chip. The IDE uses the MounRiver 2 compiler, with the option of WCH-Link for emulation and download, and provides reference examples and demonstrations of applications related to chip resources.

2. Evaluation Board Hardware

Please refer to the CH32X3x5SCH.pdf document for the schematic of the evaluation board.

CH32X3x5 Evaluation Board



Description

- | | | | |
|-----------------|-----------------|-----------------------|-----------------|
| 1、MCU | 2、USB interface | 3、DCDC | 4、 power switch |
| 5、 Reset button | 6、 SWD | 7、 Analog switch chip | |

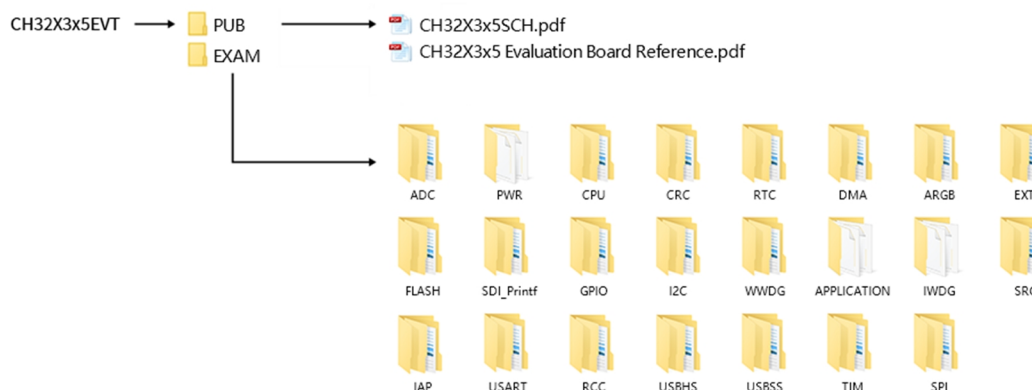
The above CH32X315 evaluation board comes with the following resources.

Motherboard - CH32X315MCU-R0

1. Main control MCU: The main control chip is CH32X315MCU6
2. USB Interface: Connects to the main chip's USBHS/USBSS communication interface
3. DCDC circuit: The buck converter chip is CH2003V, which is used to convert the 5V voltage into the 3.3V voltage that the chip can use
4. Power switch: used to cut off or connect external 5V power supply or USB power supply
5. Reset button: Connected to the IO port of the main control MCU for button control and used for external manual reset of the main control MCU
6. SWD interface: used for downloading simulation debugging
7. Analog switch chip: The multi-channel low-impedance wideband bidirectional analog switch chip CH448 achieves 8 times the number of ADC channels expansion

3. Software Development

3.1 EVT Package Directory Structure



Description:

PUB folder: provides evaluation board manuals, evaluation board schematics.

EXAM folder: Provides software development drivers and corresponding examples for the CH32X3x5 controller, grouped by peripheral. Each type of peripheral folder contains one or more functional application routines folders.

3.2 IDE Use-MounRiver

Download MounRiver_Studio, double click to install it, and you can use it after installation. (MounRiver_Studio instructions are available at the path: MounRiver\MounRiver_Studio\ MounRiver_Help.pdf and MounRiver_ToolbarHelp.pdf)

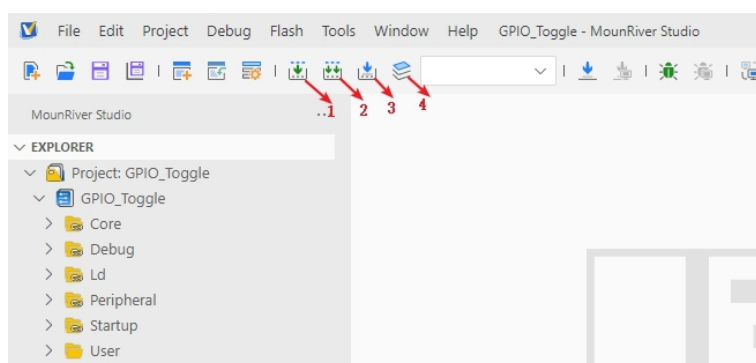
3.2.1 Open Project

➤ Open project:

- 1) Double-click project file directly with the suffix name .wvproj under the corresponding project path.
- 2) Click File in MounRiver IDE2, click Load Project, select the .project file under the corresponding path, and click Confirm to apply it.

3.2.2 Compilation

MounRiver contains three compilation options, as shown in the following figure.



Compile option 1 is Incremental Build, which compiles the modified parts of the selected project.

Compile option 2 is ReBuild, which performs a global compilation of the selected project.

Compile option 3 is incremental compilation and download, which compiles the modified parts of the selected project and then downloads them;

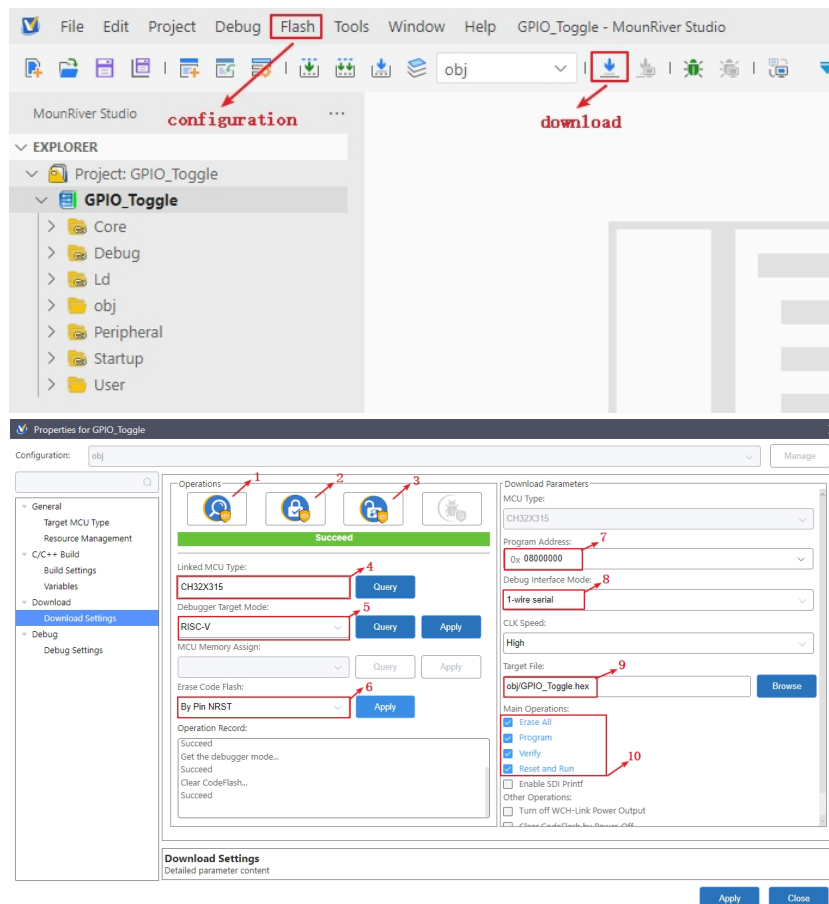
Compile option 4 is All Build, which performs global compilation for all projects.

3.2.3 Download/Simulation

➤ Download

1) Debugger download

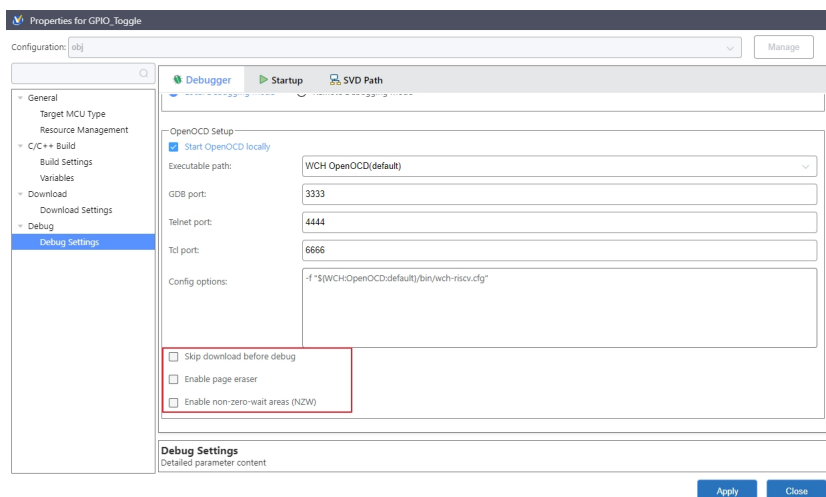
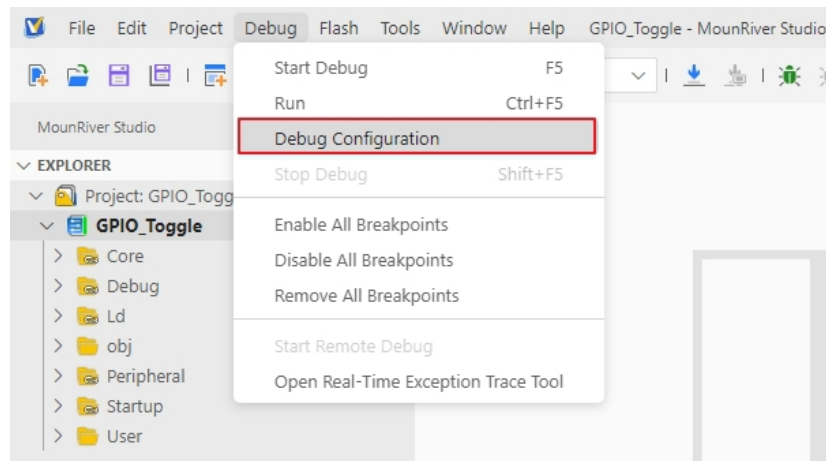
Connect to the hardware via WCH-Link (see WCH-Link instructions for details, path: MounRiver\MounRiver_Studio\ WCH-Link instructions.pdf), click the Download button on the IDE, and select Download in the pop-up interface, as shown in the figure below.



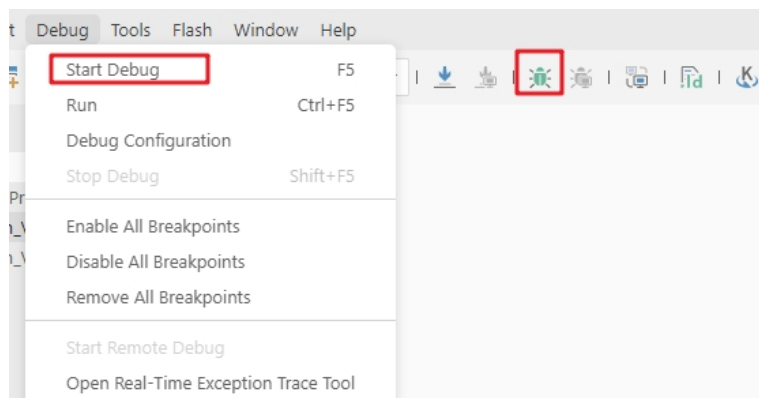
- 1 for querying the chip read protection status;
- 2 for setting the chip read protection, re-power on the configuration to take effect;
- 3 for lifting the chip read protection, re-power on the configuration to take effect;
- 4 for query the display of the current chip type
- 5 for setting the LINK mode
- 6 power-off to erase the whole user area flash
- 7 for setting the download address
- 8 for selecting the single and dual line mode
- 9 for selecting the download file
- 10 for download the configuration options

➤ Simulation

Open MounRiver Studio software for debugging configuration



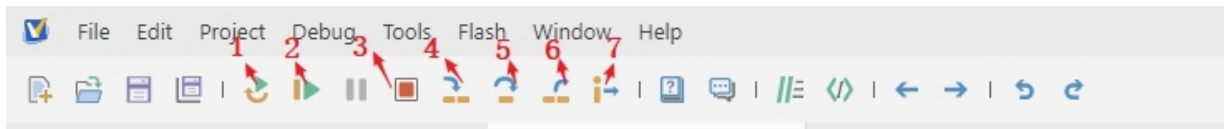
Select the appropriate debugging options, apply the settings, and save them



Click Start Debug or the debug icon to start debugging.

1) Toolbar description

Click Debug button in the menu bar to enter the download, see the image below, the download toolbar.

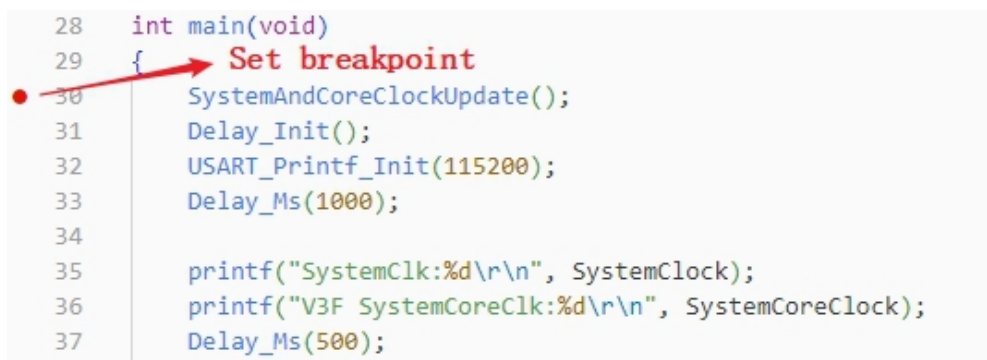


Detailed functions are as follows.

- (1) Reset: After reset, the program returns to the very beginning.
- (2) Continue: Click to continue debugging.
- (3) Terminate: Click to exit debugging.
- (4) Single-step jump-in: Each time you tap a key, the program runs one step and encounters a function to enter and execute.
- (5) Single-step skip: jump out of the function and prepare the next statement.
- (6) Single-step return: return the function you jumped into
- (7) Instruction set single-step mode: click to enter instruction set debugging (need to use with 4, 5 and 6 functions).

2) Set breakpoints

Double-click on the left side of the code to set a breakpoint, double click again to cancel the breakpoint, set the breakpoint as shown in the following figure;



3) Interface display

(1) Instruction set interface

Click on the instruction set single-step debugging can enter the instruction debugging, to single-step jump in for example, click once to run once, the running cursor will move to view the program running, the instruction set interface is shown as follows.



(2) Program running interface

It can be used with instruction set single-step debugging, still take single-step jumping in as an example, click once to run once, the running cursor will move to view the program running, the program running interface is shown as follows.

```

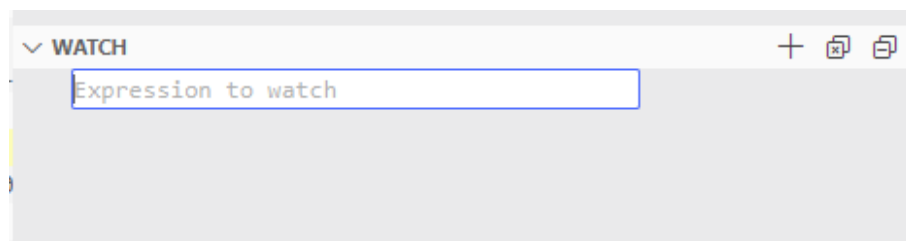
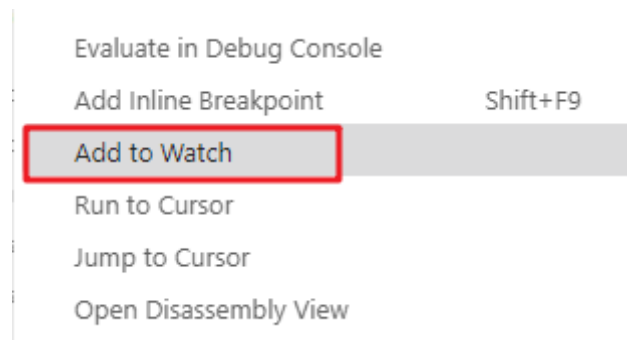
46  *
47  * @return none
48  */
49 int main(void)
50 {
51     u8 i = 0;
52
53     NVIC_PriorityGroupConfig(NVIC_PriorityGroup_2);
54     Delay_Init();
55     USART_Printf_Init(115200);
56     printf("SystemClk:%d\r\n", SystemCoreClock);
57
58     printf("GPIO Toggle TEST\r\n");
59     GPIO_Toggle_INIT();
60
61     while(1)
62     {
63         Delay_Ms(250);
64         GPIO_WriteBit(GPIOD, GPIO_Pin_0, (i == 0) ? (i = Bit_SET) : (i = Bit_RESET));
65     }
66 }
67

```

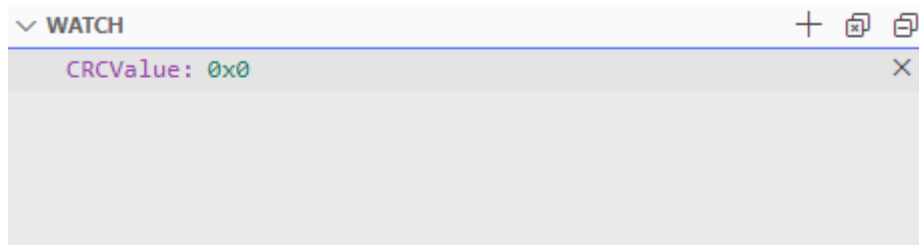
Running cursor

4) Variables

Hover over the variable in the source code to display the details, or select the variable and right-click add to watch



Fill in the variable name, and add the variable you just selected to the window:



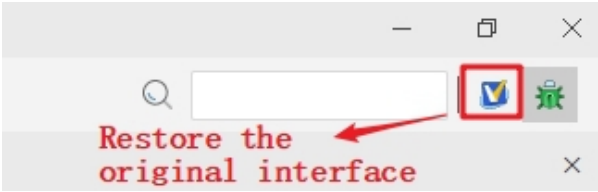
5) Peripheral registers

In the lower left corner of IDE interface Peripherals interface shows a list of peripherals, tick the peripherals will display its specific register name, address, value in the Memory window.

Problems	Output	Debug Console	Terminal	Memory	Peripheral
ADC1 Analog to digital converter					
ADC2 Analog to digital converter					
AFIO Alternate function I/O					
CAN1 Controller area network					
CAN2 Controller area network					
CAN3 Controller area network					
CRC CRC calculation unit					
DAC Digital to analog converter					

Register	Address	Value
✓ CRC	0x40023000	
> DATAR	0x40023000	0x00000000
> IDATAR	0x40023004	0x00000000
> CTLR	0x40023008	0x00000000

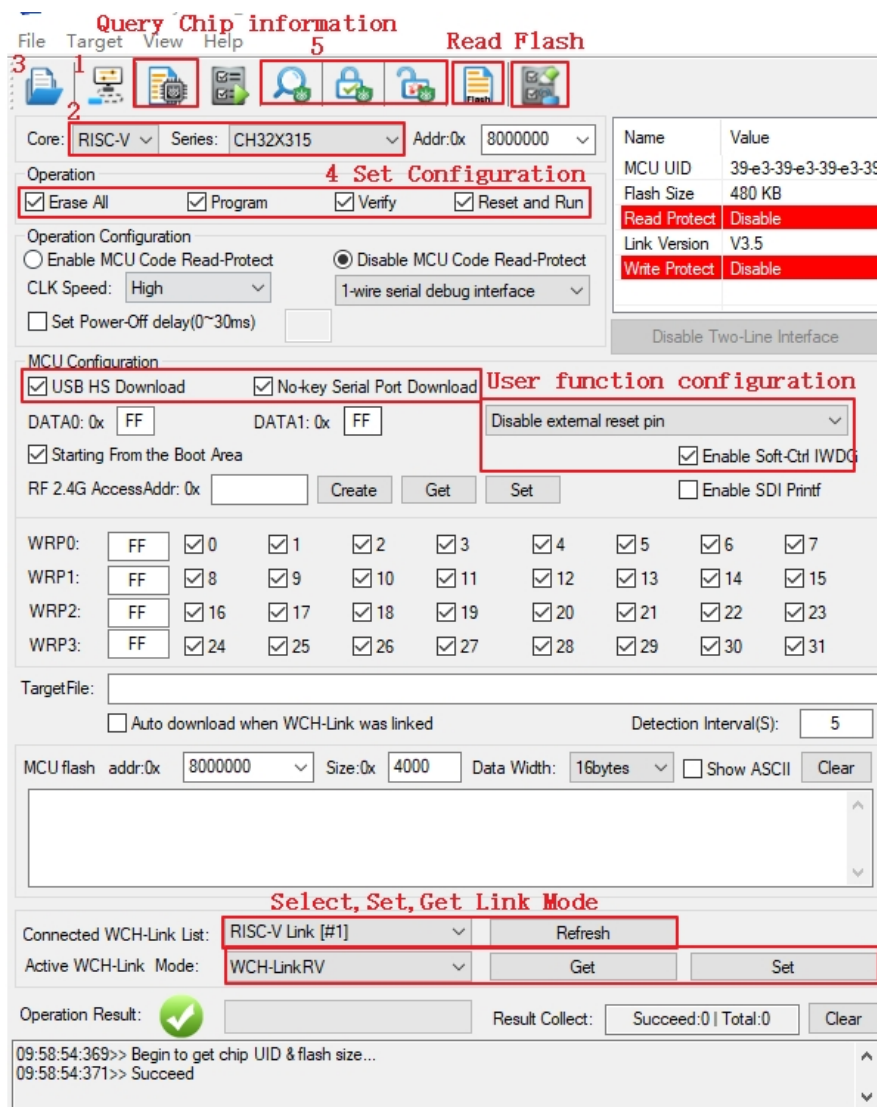
Tips: (1) When debugging, click the icon in the upper right corner to enter the original interface.



4. WCH-LinkUtility.exe Download

The download process for the chip using the WCH-LinkUtility tool is:

- 1) Connect WCH-Link
- 2) Select chip information
- 3) Add firmware
- 4) If the chip is read protected, you need to release the chip read protection.
- 5) Execute



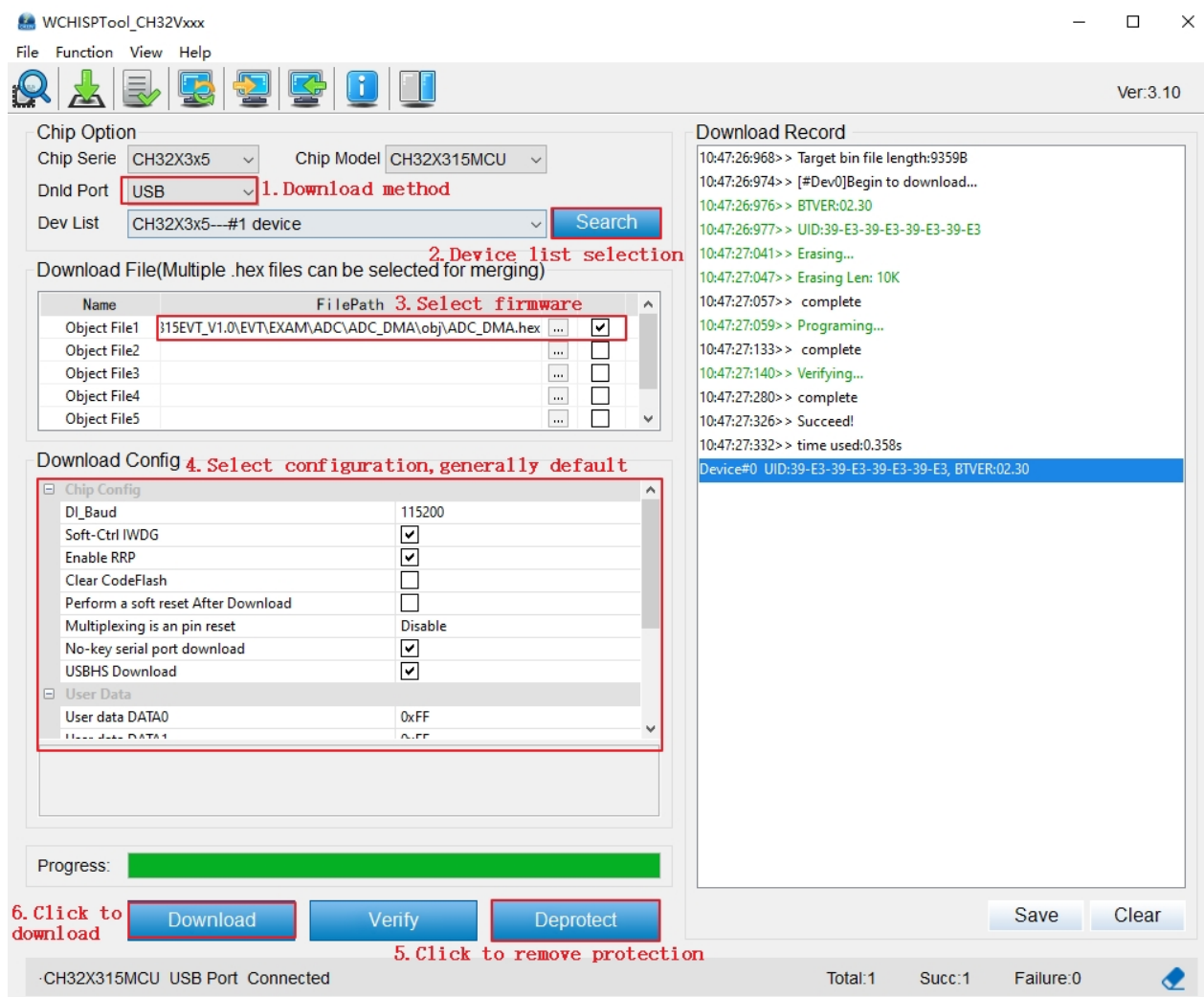
Tips: In the current version of the tool, you can configure the download URL when downloading target files.

5. WCHISPTool.exe Download

Chips now support downloading via the WCHISPTool. Both USB and serial port download methods are supported. USB pins are PD6 (USBHS_DM) and PD7 (USBHS_DP), while serial port pins are PA4 (TX) and PA5 (RX).

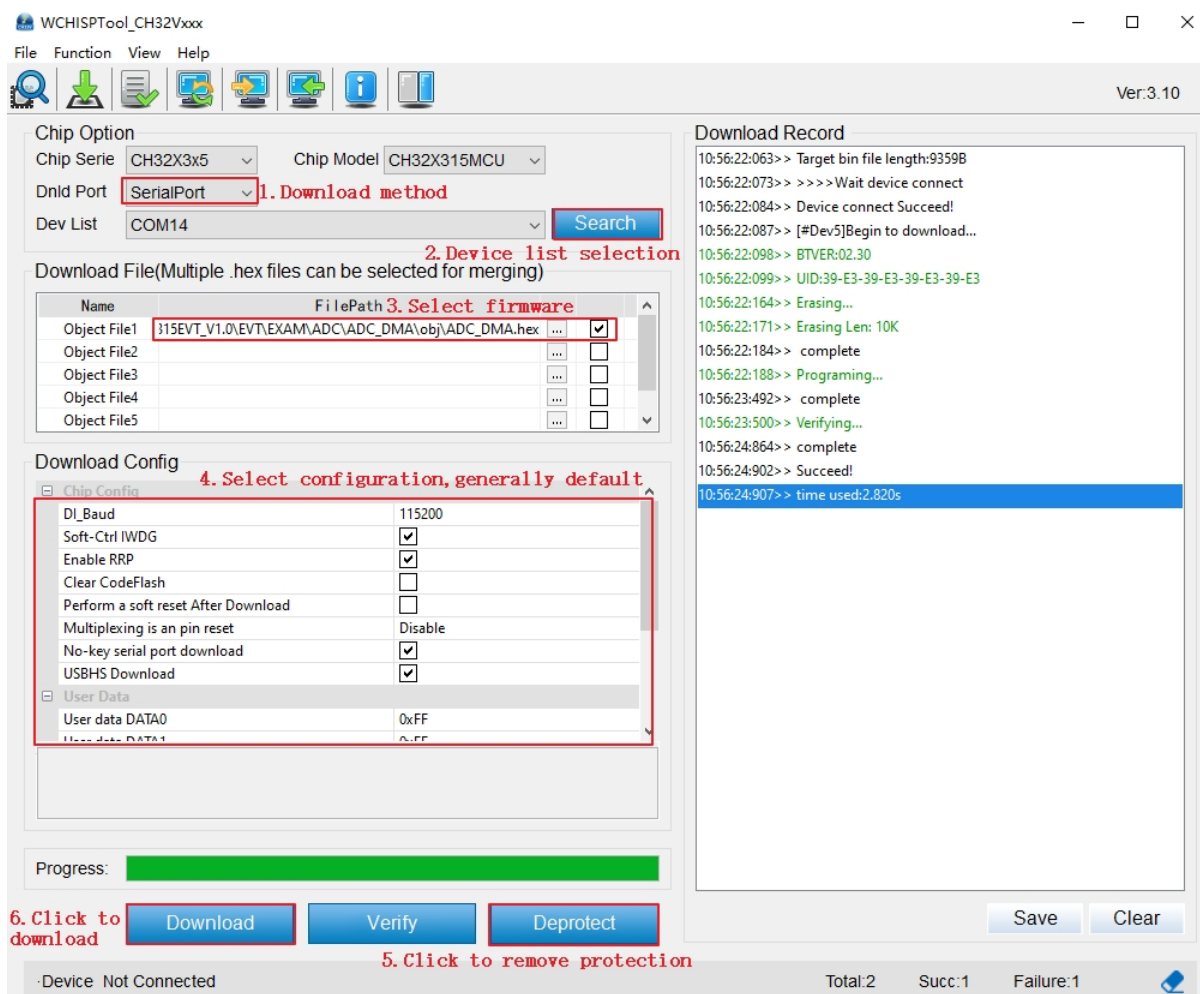
The download process is as follows:

1) USB download method



1. Select the MCU series and chip model;
2. Select the USB download method; the chip will power up again, detect a high level on the DP pin, and enter BOOT mode;
3. Identify the device; it is typically recognized automatically. If not detected, manual selection is required;
4. Select the firmware; choose the downloaded .hex or .bin target program file;
5. Select the download configuration as required;
6. Click Download.

2) Serial port download method



1. Select the MCU series and chip model;
2. Select the serial port download method and select the correct port number;
3. Select the firmware, select the downloaded .hex or .bin target program file;
4. Select download configurations as required;
5. Click Download. The software enters download-ready status, awaiting device reconnection.
6. Power cycle the device to enter BOOT mode, initiating automatic download.

Tips:

1. When using USB download, power-on check requires DP pin high and DM pin low. However, DP pin must not remain pulled up, otherwise USB functionality will be abnormal.
2. Serial port download uses a key-free method: first enable the feature, then power the device back on to enter BOOT mode for the download function to take effect. Outside BOOT mode, the verification and code protection removal processes are identical to download.
3. The key-free download feature requires enabling the corresponding option in the configuration settings; otherwise, it will not function.
4. The BOOT function is associated with corresponding user-defined function bits, both enabled by default at factory settings. These can be modified during ISP download or via a 2-wire method.
5. In blank chip state (default value in the first 4K of FLASH), when the corresponding user-defined function bit is enabled, the chip enters BOOT mode for subsequent download upon power-on without requiring additional conditions (applicable to both USB and serial interfaces).

6. Related Links

WCH Electronic Community: <https://www.wch.cn/bbs/forum-106-1.html>

WCH official website: <https://www.wch-ic.com/>

WCH-LINK instructions for use: <https://www.wch-ic.com/products/WCH-Link.html>